K-Fire WDT Solver

Water Delivery Time (WDT) Engineering Overview

Item	Description
Document Number	KF-EO-WDT-001
Document Version	1.2.1
Corresponding App Version	4.4.3
Publication Status	For Distribution
Last Updated	2026-07-10
Document Type	Applied Engineering Document for Distribution Package
Scope	Water Delivery Time (WDT) calculation principles, input and output interpretation, and applicability limits for dry-pipe and preaction systems
Related Document	K-Fire Hydraulic Calculation Engineering Overview

Revision History

Document Version	Date	Corresponding App	Change Summary
1.0.0	2026-07-10	4.4.3	Initial Engineering Overview for distribution package
1.1.0	2026-07-10	4.4.3	Adjusted public disclosure level, expanded standard gas-discharge equations, and added figure placeholders
1.2.0	2026-07-10	4.4.3	Inserted six figures and finalized document for distribution
1.2.1	2026-07-10	4.4.3	Final review: refined wording, related-document references, and terminology

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0. Introduction

0.1 Purpose and Scope

This document explains the engineering principles underlying the Water Delivery Time (WDT) function of K-Fire Hydraulic Solver. Whereas the user manual covers screen operation, this document provides the physical basis and scope of analysis needed to review calculation results or explain them to third parties.

Water Delivery Time is the time required for water to reach an open head after a sprinkler operates or a demand for water discharge occurs in a dry-pipe or preaction water-based fire protection system. Unlike steady-state hydraulic calculations for wet-pipe systems, WDT addresses air discharge from the piping, dry-side filling, water-front movement, residual-air compression, and pump and water-supply capacity together.

Category	Inclusion	Description
Dry-Pipe Valve WDT	Included	Air discharge through open heads, trip, and water transport after valve opening
Double-Interlock Preaction WDT	Included	WDT accounting for trip conditions and water transport
Single-Interlock Pre-Fill	Included	Scenario in which the secondary side is pre-filled before sprinkler operation
Arrival Time by Head	Included	Arrival time for each open head and system WDT based on the last head reached
Stabilization Time	Reference Value	Additional time from water arrival to minimum discharge pressure; separate from WDT acceptance
Steady-State Hydraulic Calculation	Reference	Pump, water supply, friction, and discharge operating points are within the scope of the hydraulic calculation document
Water Hammer and Pressure Waves	Excluded	Water hammer and elastic pressure waves are not analyzed directly
Replacement for Regulatory Determination	Excluded	Supporting information for design and review; does not replace permitting or supervisory determinations

0.2 Applicable Standards and Engineering Models

Category	Source	Role in This Document
Applicable Standard	NFPA 13 requirements related to Water Delivery Time	Basis for the acceptance concept and application of the required time
Related Function	K-Fire steady-state hydraulic calculation	Water-side friction, pump, water-supply, and device characteristics

External Engine	EPANET / WNTR	Steady-state analysis of water-side flow and pressure during filling
Engineering Theory	Compressible gas flow through an orifice and the ideal gas law	Physical basis for air discharge and residual air pressure

References in this document to “NFPA 13 criteria” concern the acceptance concept for Water Delivery Time and the application of the required time. The air-discharge model, gas laws, and EPANET hydraulic analysis are not themselves NFPA provisions; they are engineering models used by K-Fire to calculate the water-delivery process.

0.3 Key Terms

Term	Meaning
WDT	Water Delivery Time: trip time + delay time + transit time
Trip Time	Time from sprinkler operation until air pressure reaches the valve-opening condition
Transit Time	Time for water to travel from valve opening to an open head
Stabilization Time	Additional time from water arrival to minimum discharge pressure; reference value for result review
DPV	Dry-pipe valve or preaction valve; boundary between the wet side and dry side
Fill Front	Boundary between water and air in partially filled piping

In this document, “Water Delivery Time” and “WDT” are used interchangeably.

0.4 Unit System

User inputs and reports use units familiar in fire protection design practice (bar, L/min, mm, etc.), which are converted to SI units for calculations. Because the air-flow equations use absolute pressure, particular care must be taken to distinguish it from gauge pressure [bar(g)] shown on screens and in reports.

1. Overview

1.1 Purpose of Water Delivery Time

The WDT calculation answers the following questions:

- When does the dry-pipe valve or preaction valve open after a sprinkler operates?
- Along which paths does water propagate after the valve opens, and when does it reach each open head?
- Does the WDT, based on the last head reached, satisfy the project-required time?
- What is the stabilization time from water arrival until minimum discharge pressure is reached?

Under normal conditions, the piping downstream of the valve in a dry-pipe system is filled with compressed air. Therefore, steady-state discharge flow alone is insufficient; air discharge, valve opening, pipe filling, and fill-front movement must be considered together.

1.2 Relationship Between WDT and Steady-State Hydraulic Calculations

Steady-state hydraulic calculations determine how flow and pressure are distributed when the piping network is full of water and operating at a stable point. WDT calculates how a piping network not yet filled with water fills over time.

The two functions are distinct but not completely separate. Steady-state hydraulic analysis determines the instantaneous flow and pressure in water-filled sections, while the WDT calculation manages the air state, filling progress, and arrival at the heads.

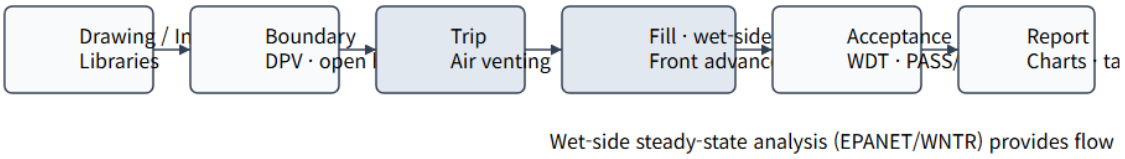


Figure 1-1. WDT Calculation Overview

1.3 Time Components

Item	Meaning	Included in WDT
Trip Time	Time for air pressure to reach the valve-opening condition	Included
Valve-Opening Delay	Input delay from trip or detection until actual valve opening	Included
Pump-Start Delay	Input delay until the pump reaches the supply condition	Included
Transit Time	Time for water to reach an open head after valve opening	Included
Water Delivery Time (WDT)	Trip + delay + transit	Acceptance Criterion
Stabilization Time	Additional time from water arrival to minimum discharge pressure	Excluded (Reference)

$$T_{\mathrm{WDT}} = t_{\mathrm{trip}} + t_{\mathrm{delay}} + t_{\mathrm{transit}} \qquad (1-1)$$

Stabilization time is useful for reviewing results but is not included in the Water Delivery Time acceptance judgment.

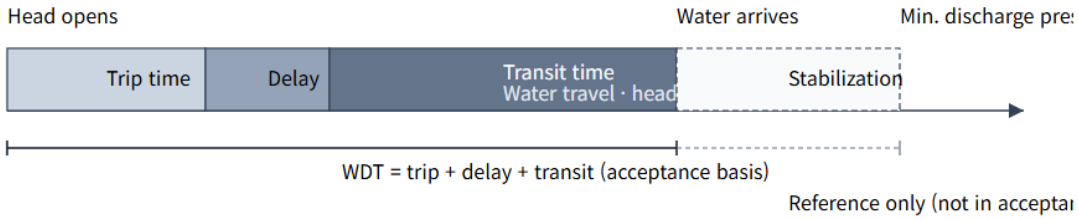


Figure 1-2. WDT Time Components

1.4 Supported Analysis Types

Analysis Type	Purpose	Key Features
Dry-Pipe Valve WDT	Trip + water delivery for a conventional dry-pipe system	Trip pressure determined from primary-side water pressure and differential ratio
Double-Interlock Preaction WDT	Trip + water delivery including detection and valve conditions	Direct trip-pressure input available
Dry-Pipe Valve + Accelerator	Evaluation of accelerator conditions such as QOD	Accounts for both trip timing and residual air pressure
Single-Interlock Pre-Fill	Secondary-side pre-fill before sprinkler operation	No trip time; subsequent delivery is evaluated from the filled state

1.5 Acceptance Criteria

When a required time is entered, pass or fail is determined using the following relationship.

$$T_{\mathrm{WDT}} \leq T_{\mathrm{required}} (1-2)$$

Acceptance	Meaning
PASS	WDT is at or below the required time
FAIL	WDT exceeds the required time
INFO	Required time not entered; calculation value is provided, but pass/fail judgment is withheld
STALL	Some or all heads not reached because fill-front advancement stopped
TIMEOUT	Delivery not completed within the calculation time limit

Acceptance is based on the last head reached. Even if some heads are reached quickly, all selected open heads must be reached within the required time.

2. Engineering Fundamentals

2.1 Physical Phenomena in Dry-Pipe and Preaction Systems

Under normal conditions, the primary side (water supply and pump to the valve) is filled with water, while the secondary side (downstream of the valve) is filled with compressed air or nitrogen. When a sprinkler operates, air is discharged; when secondary-side air pressure falls to the trip condition, the valve opens and water enters.

1. Air discharge: Air is discharged through the open head, reducing secondary-side air pressure.
2. Valve opening: The air pressure reaches the trip threshold pressure and the valve opens.
3. Filling: Water advances while filling the downstream piping, and the fill front reaches the open head.

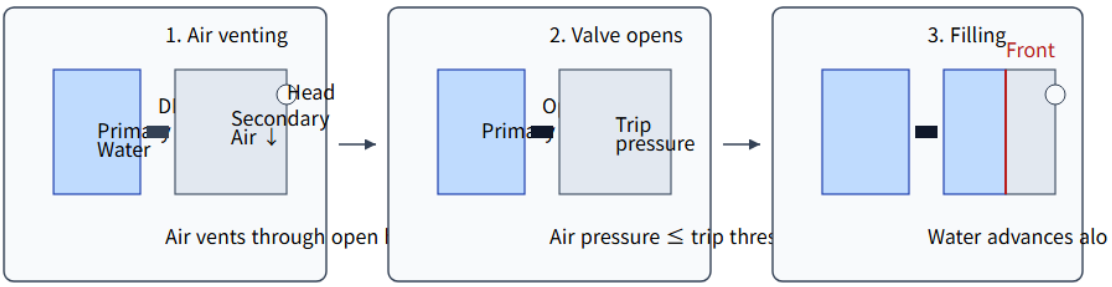


Figure 2-1. Three Stages of a Dry-Pipe System

2.2 Quasi-Steady Analysis Approach

Although water delivery is a time-dependent phenomenon, K-Fire does not use a fully transient analysis that directly solves water hammer or elastic pressure waves. At each time step, the water-filled region is assumed to be instantaneously near steady state, and EPANET/WNTR calculates the flow and pressure at that time.

Phenomenon	Handled By
Water-side friction, pump curve, water supply, and devices	EPANET/WNTR steady-state analysis
Air discharge and air-pressure change	Compressible-gas model
Filling progress and head arrival	WDT time analysis

This quasi-steady approach is a simplification of a fully compressible two-phase transient analysis. However, the effects of pipe length, diameter, elevation, pump curve, equivalent length, and valve status from the project drawing on water transport are represented using the actual model.

2.3 Coupling of Compressible Gas and Incompressible Water

Water is treated as incompressible, with flow and pressure determined by steady-state hydraulic analysis. Air is treated as compressible, with mass and pressure modeled using ideal-gas relationships and an orifice-discharge model.

Pressure in the residual-air region is represented by the ideal gas law.

$$P_{\text{air}} = \frac{m_{\text{air}} R_s T}{V_{\text{air}}} \qquad (2-1)$$

Symbol	Meaning
P_{air}	Residual-air absolute pressure
m_{air}	Residual-air mass
R_s	Gas constant
T	Absolute temperature
V_{air}	Residual-air volume

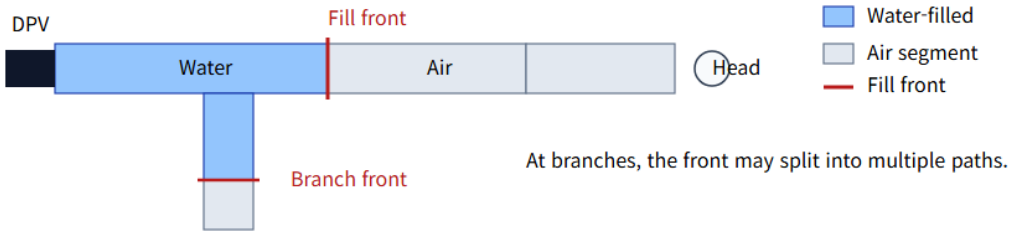
As water advances, V_air decreases; as air vents through open heads, m_air also decreases. Which effect dominates depends on the piping network and head conditions. Therefore WDT cannot be computed simply as “dry-side volume ÷ pump flow.”

2.4 Fill-Front Concept

The fill front is the boundary between water-filled sections and sections that still contain air. Because actual sprinkler piping networks have many branches and junctions, the front may divide in multiple directions or merge with other fronts.

Steady-state hydraulic calculations alone cannot determine which pipes filled and when. WDT links water-side flow results to filling progress and head arrival times.

Figure 2-2. Fill-Front Concept



3. Pre-Trip Air Discharge

3.1 Trip Threshold Pressure

A dry-pipe valve opens when secondary-side air pressure has decreased sufficiently. The trip threshold pressure of a dry-pipe valve is generally determined from primary-side water pressure and the valve differential ratio.

$$P_{\text{trip}} = \frac{P_{\text{primary}}}{R_{\text{diff}}} \qquad (3-1)$$

Symbol	Meaning
P_{trip}	Trip threshold air pressure
P_{primary}	Valve primary-side water pressure

R_{diff}	Valve differential ratio

For double-interlock preaction systems, the trip pressure can be entered directly. If the initial air pressure is already at or below the trip pressure, trip time may be treated as zero; if discharge to atmosphere alone cannot reach the trip pressure, the result is judged as failure to trip.

3.2 Standard Gas-Discharge Equations

An open head acts as an orifice for air discharge. The discharge rate depends on open area, upstream air pressure, gas type, temperature, and discharge coefficient. Water-discharge relations such as Q = KVP are not applied directly to air.

Taking the specific-heat ratio of air or nitrogen as κ = 1.40, choking under atmospheric discharge is judged with the following critical pressure ratio.

$$\frac{P_b}{P_0} \leq \left(\frac{2}{\kappa+1}\right)^{\frac{\kappa}{\kappa-1}} \approx 0.528 \qquad (3-2)$$

When upstream absolute pressure is sufficiently high for choked conditions, the standard choked mass-flow equation is as follows.

$$\dot{m} = C_d A P_0 \sqrt{\frac{\kappa}{R_s T}\left(\frac{2}{\kappa+1}\right)^{\frac{\kappa+1}{\kappa-1}}} \qquad (3-3)$$

When the pressure decreases to subsonic conditions, the standard subsonic mass-flow equation is as follows.

$$\dot{m} = C_d A P_0 \sqrt{\frac{2\kappa}{R_s T(\kappa-1)}\left[\left(\frac{P_b}{P_0}\right)^{\frac{2}{\kappa}} - \left(\frac{P_b}{P_0}\right)^{\frac{\kappa+1}{\kappa}}\right]} \qquad (3-4)$$

Symbol	Meaning
$\dot{m}$	Air mass flow rate
C_d	Air-discharge coefficient
A	Orifice area
P_0	Upstream-air absolute pressure
P_b	Downstream (atmospheric) absolute pressure
κ	Specific heat ratio
R_s	Gas constant
T	Absolute temperature

Equations (3-2) through (3-4) are standard engineering equations for compressible orifice flow. The method of determining vent area from the head K-Factor and applying the discharge coefficient follows product input and library policies.

3.3 Thermodynamic Assumptions

Because pre-trip air discharge occurs over a short period, either an adiabatic or isothermal assumption may be selected. The product's default calculation path uses an adiabatic model, which more closely represents rapid discharge. Actual installations may experience temperature changes, heat transfer, leakage, and localized air pockets; therefore, input sensitivity and limitations must also be reviewed.

3.4 Accelerator (QOD)

When an accelerator is installed, the valve may open sooner without waiting for the normal trip threshold pressure. Accounting for an accelerator is not simply a matter of subtracting a fixed time from the result. Because the valve may open at a higher residual air pressure, both the trip time and the residual air pressure at the start of transit are represented.

4. Dry-Side Volume and Boundaries

4.1 Wet-Side/Dry-Side Boundary

The first determination in WDT is which piping is normally filled with water and which is filled with air. K-Fire treats the DPV as the boundary and divides the wet side from the dry side based on connectivity.

Area	Definition	Role
Wet Side	Water-filled area between the water supply/pump and DPV	Water-side supply path
Dry Side	Air-filled area downstream of the DPV and connected to open heads	Trip volume and filling target
Isolated Area	Area inaccessible to air or water because of a closed valve or similar isolation	Excluded from calculation or flagged with a warning

The dry side is not simply “all piping with sprinklers.” The analysis considers whether air communicates with an open head, whether the piping is downstream of the valve, and whether it is isolated by a closed valve.

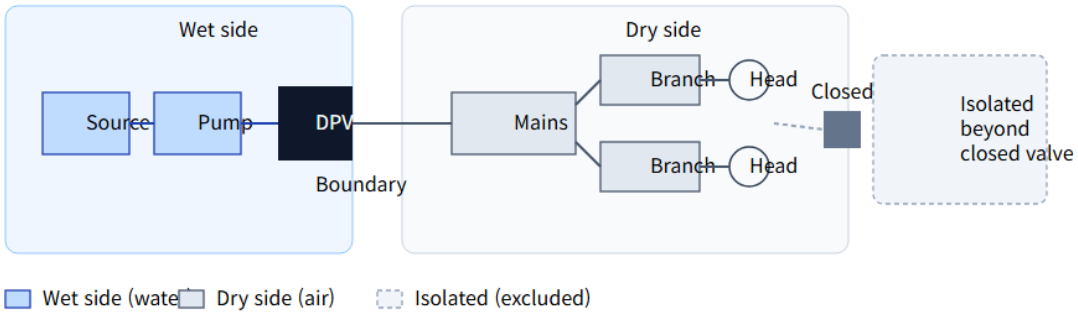


Figure 4-1. Wet-Side/Dry-Side Boundary

4.2 Dry-Side Volume

The dry-side air volume is calculated as the sum of the internal volumes of the applicable piping.

$$V_{\text{air,total}} = \sum_i \frac{\pi}{4} D_i^2 L_i \qquad (4-1)$$

Symbol	Meaning
$V_{\text{air,total}}$	Initial dry-side air volume
D_i	Inside diameter of pipe
L_i	Length of pipe

If volume cannot be calculated because pipe diameter or length is missing, the analysis stops. Dead-end piping is included in the volume if it is within an air-connected area.

4.3 Open Heads and Unreachable Areas

An open head is both the pre-trip air outlet and the path through which air ahead of the fill front escapes during transit. A closed valve can interrupt air and water communication; if a bypass path exists, the area may still be reached through another route.

Only one active DPV is permitted in a single calculation. If an open head is not on the dry side or the DPV is ambiguous, the analysis stops and requires input correction.

5. Water Transport and Fill Front

5.1 Transit Initial Conditions

Transit begins when the valve opens and water enters the dry side. Under normal dry-pipe conditions, the air pressure at the start of transit equals the trip threshold pressure. With an accelerator, transit may begin at a higher residual air pressure.

A conventional dry-pipe WDT calculation begins with an unfilled dry side, whereas transit following single-interlock pre-fill may begin with some piping already filled.

5.2 Front Advancement and Branching

When water enters the dry side, the fill front advances. If the piping network branches, the front may divide among multiple paths; when water arriving from different paths meets, that section is considered filled.

Air ahead of the fill front is vented if connected to an open head; if not connected, it may remain as a trapped air pocket. This distinction is based on piping connectivity.

The basic relationship for fill-front movement is expressed in terms of inflow and pipe cross-sectional area.

$$\Delta x = \frac{Q \Delta t}{A} \qquad (5-1)$$

Symbol	Meaning
Δx	Fill-front travel distance
Q	Water inflow to the applicable front
Δt	Time interval
A	Pipe cross-sectional area

Equation (5-1) is the fundamental volume-conservation relationship. The actual analysis also accounts for piping-network topology, air backpressure, and supply capacity.

5.3 Transition to Water Discharge After Head Arrival

When the fill front reaches an open head, the head transitions from an air outlet to a water-discharge outlet. Subsequent discharge flow is handled in the same form as in a conventional hydraulic calculation.

$$Q = K\sqrt{P} \qquad (5-2)$$

Symbol	Meaning
Q	Head discharge flow
K	Head K-Factor
P	Head pressure

WDT acceptance is based on the time water arrives, not the time the head reaches minimum discharge pressure. Stabilization time is recorded separately as a reference value.

6. Role of Steady-State Hydraulic Analysis

During a WDT calculation, EPANET/WNTR performs the water-side steady-state analysis. It is used to determine flow and pressure in the currently water-filled region and the inflow entering the fill front.

Category	Steady-State Hydraulic Calculation	Role in WDT
Purpose	Final operating point itself	Intermediate water-side flow and pressure during filling
Model Scope	Entire water-filled system	Reflects the current extent of water delivery
Air and Fill Front	Not addressed	Managed by WDT time analysis

The general principles for friction equations (H-W/D-W), pump curves, water supplies, and head-discharge characteristics follow the hydraulic calculation document. Because WDT results reflect filling progress, they should not be compared directly on a one-to-one basis with the final operating point from a conventional hydraulic calculation.

Boundary errors, failure to trip, fill-front stall, and time limits are distinguished by stage and displayed to the user. Numerical results may be produced even when warnings are present, but the designer must investigate their causes.

7. Input Data

WDT accuracy is directly affected by drawing inputs, libraries, trip and air conditions, supply capacity, and analysis options.

7.1 Drawing and Boundary Inputs

Input	Primary Effect
DPV or preaction-valve location	Dry-side volume and active system
List of open heads	Trip time, transit paths, and heads subject to acceptance
Pipe length and inside diameter	Volume, filling time, and friction loss
Pipe elevation	Supply pressure, stratification risk, and stabilization
Closed valves	Unreachable and isolated areas
Piping connectivity	Boundary errors and potential failure to reach

7.2 Trip and Air Inputs

Input	Description
Analysis Type	Dry-pipe, double-interlock, and single-interlock pre-fill
Primary-side water pressure and differential ratio	Dry-pipe valve trip-pressure calculation
Direct Trip Pressure	Used for double interlock
Initial Air Pressure	Starting pressure before trip
Gas Type and Temperature	Air or nitrogen; determination of state variables
Air-discharge coefficient	Correction of orifice discharge rate
Thermodynamic Mode	Adiabatic or isothermal
Accelerator Setting	Trip correction for QOD or similar accelerator

7.3 Supply and Hydraulic Inputs

The transit stage uses the same water-side inputs as a steady-state hydraulic calculation. Pump curves, water supplies, friction coefficients, equivalent lengths, head K-Factors, and minimum discharge pressure affect filling flow and stabilization time.

Results are particularly sensitive to pipe volume, initial air pressure, air-discharge coefficient, open-head locations, and pump supply capacity.

7.4 Analysis Options

Option	Meaning
Required Time	PASS/FAIL acceptance criterion
Valve-opening delay and pump-start delay	Input delays included in WDT
Open-Head Restriction	Designates only specified heads as subject to acceptance
Default Minimum Discharge Pressure	Used to calculate stabilization time
Pre-Fill Options	Pre-fill termination and subsequent-transit conditions

8. Outputs and Result Interpretation

Results are evaluated based on the last head reached among all selected open heads.

8.1 Summary Results

Item	Meaning	Use in Acceptance
Trip Time	Result of the air-discharge stage	WDT component
Delay Time	Sum of valve and pump delays	WDT component
Transit Time	Based on the last head reached	WDT component
Water Delivery Time (WDT)	Trip + delay + transit	PASS/FAIL criterion
Required Time	User input	Acceptance criterion value
Time Margin	Required time − WDT	Positive value indicates available margin
Stabilization Time	From WDT to minimum discharge pressure	Reference Value
Acceptance	PASS, FAIL, INFO, STALL, etc.	Result acceptance

8.2 Results by Head

Arrival time by head is a key output for confirming the arrival sequence and the last head reached. If any head is not reached, the entire system is treated as not reached.

8.3 Graphs, Playback, and Reports

WDT results include time histories as well as a single numerical value.

- Pre-trip air-pressure decay
- Filling progress by pipe
- Pressure and flow at pump, valve, and head monitoring points
- Pre-fill fill-ratio and residual-air-pressure curves

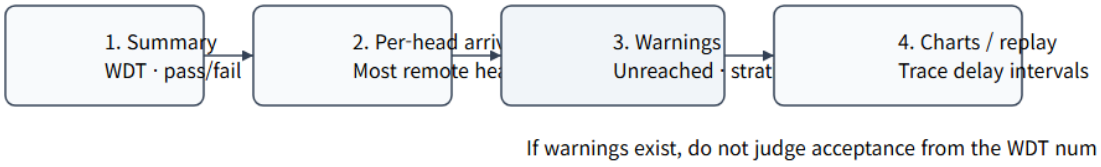


Figure 8-1. Result Interpretation Flow

Recommended report review sequence:

4. Analysis type, active DPV, and list of open heads
5. Relationship of trip, delay, transit, and WDT to the required time
6. Last head reached and whether any heads were not reached
7. Review stabilization time separately from WDT acceptance
8. Review warnings, stratification risk, and the pre-fill disclaimer

9. If necessary, use filling playback and monitoring-point graphs to trace delayed sections

8.4 Warning Interpretation

A warning does not necessarily indicate calculation failure. It identifies a condition that the designer must review even if the calculation was completed.

Category	Meaning
Boundary Warning	DPV, closed valves, and unreachable areas
Input Warning	Missing required pressure, K-value, or pipe properties
Trip Warning	Trip condition cannot be reached or conditions are unrealistic
Transit Warning	Fill-front stall, time limit, or heads not reached
Stratification Risk	Potential partial filling of horizontal, large-diameter piping
Pre-Fill Warning	Unable to fill, time limit, etc.

A result with warnings must not be deemed acceptable solely on the basis of the WDT value.

9. Single-Interlock Pre-Fill

In a single-interlock preaction system, a detection signal opens the valve, allowing the secondary side to be pre-filled before a sprinkler actually operates.

9.1 Purpose and Sequence

Pre-fill estimates how far the secondary side fills and the value toward which residual air pressure converges when the valve opens first while the heads remain closed. This result can be used as the initial condition for subsequent post-fill transit.

Item	Conventional Dry-Pipe WDT	Single-Interlock Pre-Fill
Initiating Event	Sprinkler operation	Valve opening by detection and control
Trip Stage	Trip by air discharge	No trip time (or external control)
Head Status	Open head serves as air outlet	Heads remain closed during pre-fill
Primary Result	WDT = trip + delay + transit	Fill time, fill ratio, and post-fill transit

9.2 Analysis Considerations

During the pre-fill stage, the heads are closed, so air is assumed not to discharge to atmosphere. As water enters, the residual-air volume decreases and air pressure rises in accordance with Equation (2-1). The value approached at equilibrium depends on the termination-condition settings, pump curve, initial air pressure, and pipe volume.

For post-fill transit acceptance, trip time is set to zero, and valve and pump delays are added to post-fill transit time. Pre-fill time and post-fill transit time are shown in the same report but are not automatically added together as a single WDT.

Pre-fill time is a theoretical calculation without an interface to application requirements or field measurements. Actual control logic, detection delay, valve characteristics, and field venting conditions require separate confirmation.

10. Assumptions, Limitations, and Applicability

Stating the limitations is necessary for an engineering-sound interpretation of the results.

10.1 Limitations of Quasi-Steady Analysis

The following phenomena are not analyzed directly.

Excluded Phenomenon	Description
Water Hammer and Pressure Waves	Elastic transient response due to rapid opening, closing, or startup
Detailed Two-Phase Turbulence	Slugging, spray, and local interface structures
Heat-Transfer Transients	Detailed time-dependent heat transfer among

	piping, air, and water
Air Leakage	Time-dependent leakage from joints, trim, and similar components
Detailed Control Logic	Manufacturer-specific operation of detectors, solenoids, and panels

WDT results are engineering estimates that reflect dry-side volume and supply capacity. Where field acceptance-test results, manufacturer data, or AHJ requirements are available, those sources take precedence.

10.2 Two-Phase and Stratified Flow

K-Fire models the fill front as a boundary that advances while filling the pipe cross-section. In long, horizontal, large-diameter piping, stratified flow may occur, with water flowing along the bottom and air remaining at the top. Stratification-risk diagnostics do not automatically correct the result; they issue a warning to prompt design review.

10.3 Input and Use Limitations

Pipe inside diameter, length, elevation, friction coefficient, equivalent length, head K-Factor, pump curve, and water-supply conditions all significantly affect the results. Library defaults are provided to assist with design input and must be checked against manufacturer-approved documents, field specifications, and applicable standards.

K-Fire results are engineering calculation data intended to support design and review. Regulatory permitting, supervisory approval, and final determinations of responsibility are governed by the standards applicable to the project and professional engineering review. When warnings are present, an acceptable determination must not be made solely because WDT is at or below the required time.

11. Verification Approach

K-Fire WDT verification consists of the following categories.

Category	Verification Focus
Unit-Level Physical Equations	Pressure-unit conversion, choked/subsonic transition, ideal-gas relationships, and trip-pressure calculation
Boundaries and Volume	DPV-based separation, open-head selection, closed valves, and volume calculation
Transit and Timeline	Straight runs and branches, heads not reached and time limits, stabilization time, and stratification warnings
Reports and Reproducibility	Summary and per-head tables, graphs, pre-fill disclaimer, and result reproduction
External Engine	Numerical accuracy and limitations of EPANET/WNTR are within the scope of the applicable engine documentation and license

K-Fire provides fire protection system modeling, unit conversion, air and filling-time analysis, and a reporting layer on top of a third-party hydraulic analysis engine.

Appendices

Appendix A. Key Symbols

Symbol	Meaning	Representative Unit
T_{WDT}	Water Delivery Time	s
t_{trip}	Trip Time	s
t_{delay}	Valve and pump delay time	s
t_{transit}	Transit Time	s
P_{air}	Air pressure	bar(g) / Pa(abs)
P_{trip}	Trip threshold air pressure	bar(g)
P_{primary}	Valve primary-side water pressure	bar(g)
R_{diff}	Differential ratio	-
V_{air}	Residual-air volume	m^3, L
m_{air}	Residual-air mass	kg
$\dot{m}$	Air mass flow rate	kg/s
Q	Water flow	L/min
K	Head K-Factor	L/min/√bar
C_d	Air-discharge coefficient	-
κ	Specific heat ratio	-
R_s	Gas constant	J/(kg·K)
A	Orifice or pipe cross-sectional area	m^2

Appendix B. List of Figures

Number	Title	File
Figure 1-1	WDT Calculation Overview	figures/wdt/fig_1_1_wdt_overview.svg
Figure 1-2	WDT Time Components	figures/wdt/fig_1_2_time_components.svg
Figure 2-1	Three Stages of a Dry-Pipe System	figures/wdt/fig_2_1_three_phases.svg
Figure 2-2	Fill-Front Concept	figures/wdt/fig_2_2_front_concept.svg
Figure 4-1	Wet-Side/Dry-Side Boundary	figures/wdt/fig_4_1_wet_dry_boundary.svg
Figure 8-1	Result Interpretation Flow	figures/wdt/fig_8_1_result_flow.svg

Appendix C. References

Verify cited edition, section, and table numbers against the original source.

10. NFPA 13, Standard for the Installation of Sprinkler Systems.
11. EPANET hydraulic analysis documentation.
12. WNTR documentation and license notice.
13. Compressible gas flow through orifices and nozzles.
14. Ideal gas law and thermodynamic relations for air and nitrogen.

15. Two-phase filling and stratified flow references for partially filled fire protection piping.

— End of document (v1.2.1) —